

Skills Summary

Separation and Initial-Condition Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Separate the variables before integrating

Rule: an integral cannot be evaluated while the unknown function is still on the wrong side.

$$\frac{dy}{dx} = g(x)h(y) \implies \int \frac{dy}{h(y)} = \int g(x) dx$$

Test before integrating: does the right side contain y ? If it does, move it across first. Treating y as a constant inside $\int \dots dx$ is the single most common wrong first line.

Example: Solve $\frac{dy}{dx} = 4xy$ with $y(0) = 2$ and find $y(1)$.

Step 1. The right side contains y , so nothing may be integrated until the variables are separated.

Step 2. $\frac{dy}{y} = 4x dx$ gives $\ln |y| = 2x^2 + C$, so $y = Ae^{2x^2}$; $y(0) = A = 2$, and $y(1) = 2e^2 \approx 14.78$.

Try it: Solve $\frac{dy}{dx} = 6xy$ with $y(0) = 1$ and find $y(1)$.

check 20'06

Skill 2 — Where the constant of integration goes

Rule: exponentiating a *sum* produces a *product*.

$$\ln |y| = kx + C \implies y = e^C e^{kx} = Ae^{kx}$$

So the additive $+C$ becomes the multiplying factor A . Writing $y = e^{kx} + C$ is the classic error, and it is caught in one line by differentiating: $y' = ke^{kx}$ while $ky = ke^{kx} + kC$.

Example: Solve $\frac{dy}{dx} = 3y$ with $y(0) = 4$ and find $y(1)$.

Step 1. The equation is already separable in y alone, so the whole question is where the constant lands after exponentiating.

Step 2. $\ln |y| = 3x + C$ gives $y = Ae^{3x}$; $y(0) = A = 4$, so $y = 4e^{3x}$ and $y(1) = 4e^3 \approx 80.34$.

Try it: Solve $\frac{dy}{dx} = 2y$ with $y(0) = 5$ and find $y(1)$.

check 96'98

Skill 3 — When to apply the initial condition

Rule: integrate first, solve for the constant second, simplify third. An initial condition can only be used once a constant exists to determine.

And solve for y carefully: from $y^2 = x^2 + K$ the root applies to the *whole* side, $y = \pm\sqrt{x^2 + K}$, with the sign chosen by the initial condition. $\sqrt{u^2 + v^2} \neq u + v$.

Example: Solve $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = 3$ and find $y(4)$.

Step 1. Separate and integrate before touching the initial condition; the condition then fixes the constant inside the radical, not outside it.

Step 2. $y dy = x dx$ gives $y^2 = x^2 + K$; $y(0) = 3$ gives $K = 9$, so $y = \sqrt{x^2 + 9}$ and $y(4) = \sqrt{16 + 9} = 5$.

Try it: Solve $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = 5$ and find $y(12)$.

check: 13

(!) Watch out: every particular solution can be checked in one line — differentiate it and compare with the original right-hand side. A solution that fails that test is wrong no matter how tidy the algebra looked.